E-Learning Platform

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ABSTRACT

This project presents a City Transport System implemented using graph data structures in C. Stations are vertices and connections are edges using adjacency list representation. CRUD operations are supported for efficient data handling.

Keywords— Binary Search Tree, Course Management System, Data Structures, C Programming, E-learning Platform

I. INTRODUCTION

A. Transportation systems require efficient management of connections. Graphs represent stations and routes effectively. This project uses adjacency lists for memory-efficient implementation.

II. OBJECTIVES

The main objectives of this project are:

- Implement graph using adjacency list
- Perform CRUD operations
- Use dynamic memory allocation
- Build menu-driven system

III. SCOPE OF PROJECT

B. This system models a simplified transport network and allows manual management of station connections.

IV. LITERATURE REVIEW

Graphs are widely used in transportation, social networks, and communication systems. Adjacency lists are efficient for sparse graphs and support dynamic operations effectively. These properties make them suitable for transport systems.

V. METHODOLOGY

C. Define node structure, initialize adjacency list, implement CRUD operations, and provide menu-driven interface.

VI. TECHNOLOGY USED

- Programming Language: C
- Data Structure: Graph
- Compiler: GCC
- Platform: (Windows/Linux/Mac)
- Interface: Console faced
- Input method: Standard Input(scanf)
- Output method: Console Output(printf)

VII. ARCHITECTURE/SYSTEM DESIGN

The system follows a hierarchical tree-based design:

Course
→graphs

Each station is a vertex and connections are edges stored using adjacency lists.

VIII. IMPLEMENTATION DETAILS

The program is implemented using structures for Course, Module, and Lesson. Dynamic memory allocation is used for storing strings and nodes. Key functions include:

Functions include addEdge(), deleteEdge(), updateEdge(), search(), and display().

IX. RESULTS / OUTPUT

Sample output:

```
==============================
 CITY TRANSPORT SYSTEM MENU
==============================
1. Add Station (Create)
2. Display Network (Read)
3. Update Station (Update)
4. Delete Station (Delete)
5. Search Station
6. Exit
Enter Choice:

Enter Choice: 1
Enter Station ID: 123
Enter Station Name: payinur
Enter Transport Mode (Metro/Train/Bus): bus
SUCCESS: Station added to the network!
```

X. CONCLUSION

The project demonstrates graph-based transport system implementation effectively.

XI. FUTURE WORK

Future enhancements may include:

- Graphical User Interface (GUI)
- Database integration
- Add station name
- Shortest path algorithms
- Web-based e-learning platform
- User authentication system

XII. REFERENCES

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